

## **A second dossier at Haghia Triada, and a category of personnel written with a bare sign**

**Alberto Acedo** Biome Makers Inc. Draft v0.2, 9 September 2026. The prior art of Younger, Uchitel, Davis and Valério, and Montecchi is situated throughout.

### **Abstract**

The class  $\Omega$  of Montecchi's revised classification of the Haghia Triada tablets (2019) proves, on measurement, to be organised around the sign \*305, and to constitute an administrative dossier distinct from the aromatics dossier described elsewhere. Seeding a co-occurrence analysis with \*305 over the 1,719 documents of the corpus yields a block of six members which shares only six of its twenty-four documents with the sixty-eight of the aromatics block; the two are separated most sharply by their arithmetic, the aromatics dossier recording 54% of its quantities in fractions and the \*305 dossier 5%. Much of what can be said about \*305 itself is already published and is set out as such here: Younger observes that it occurs with groups of VIR and concerns personnel; Uchitel (1994-95, 2002-03) reads the first list of HT 94a as personnel recorded by collective designations, identifies ka-pa as a Cretan toponym and sa-ra<sub>2</sub> as a designation of personnel, and notes the near-equal figures under both headings on HT 105; Davis and Valério (2020) propose that entries with figures above one may be non-anthroponymic designations of groups; and Montecchi reads the four tablets of this type as records of feast preparation. What this note contributes is measurement of four kinds. First, the arithmetic that turns Younger's observation into a result: on HT 100 the surviving figures sum to 93 against a recorded total of 97, so the lost figure would be 4, and on HT 94a the parallel sum gives 111 against 110, a discrepancy Montecchi (2009) had already identified as a scribal miscalculation and which shows that bare signs belong inside personnel totals. Second, a filter: since people are not divided, any sign carrying fractions elsewhere in the corpus cannot be a category of persons, which excludes RE, TI and DI and leaves five candidates, of which TI+A is the cleanest, occurring twice and never outside a personnel account. Third, the observation that all five surviving candidates are exclusive to Haghia Triada, so that the villa classified its personnel with signs of its own. And fourth, a datum bearing on Uchitel's identification of \*306 as the Linear A sign for woman: his reading of HT 94a rests on a parallel with the Pylian Aa and Ab series, where rations are fixed, and no fixed ration scale exists in this corpus, the figs-to-barley ratio ranging from 0.38 to 1.00 across the six tablets he tabulates. The personnel figures are not rounded, which excludes their being quotas, and \*327 is recorded at 33 and 34 on two different tablets, the only case in the corpus of the same category counted twice.

**Keywords:** Linear A, Haghia Triada, personnel, administrative dossiers, Montecchi classification, null models.

### 1. Where the question comes from

The revised classification of the Haghia Triada tablets (Montecchi 2019, pp. 97-107) introduces classes that its 2010 predecessor did not have, among them three designated with Greek letters. One of these, class  $\Omega$ , gathers seven tablets: HT 37, 41, 45, 57, 63, 81 and 144.

Working on that classification for a different purpose, I noticed that those seven tablets are organised around a single sign, \*305, and that \*305 was one of three signs I had isolated some days earlier by their association with the aromatics dossier of the same archive (Acedo 2026e), by hypergeometric co-occurrence and with no knowledge that it had been made a class. Two routes, one by inspection of content and structure and one by distributional measurement, had converged on the same group.

That convergence is the starting point of what follows.

### 2. The dossier, measured

Growing a block from \*305 by hypergeometric association over the 1,719 documents (retaining signs that co-occur at  $p < 0.001$  in two rounds) yields six members:

| sign                   | occurrences | documents | sites                | quantities                    |
|------------------------|-------------|-----------|----------------------|-------------------------------|
| <b>*305</b>            | 12          | 11        | Haghia Triada only   | 90, 23, 8, 7, 5, 3, 2, 2, 1   |
| TU                     | 11          | 11        | HT, Khania, Phaistos | 6, 6, 4, 1, 1, $\frac{1}{16}$ |
| ZU                     | 5           | 5         | HT, Khania           | 5, 3, 2                       |
| OLE+NE                 | 5           | 4         | HT, Tylissos         | 2, $\frac{3}{4}$              |
| QA <sub>2</sub> +RE+PU | 4           | 3         | Haghia Triada only   | 10, $1\frac{1}{4}$            |
| *318                   | 3           | 3         | Haghia Triada only   | (no quantity)                 |

Nine documents carry two or more members: HT 23a, 33, 45a, 45b, 63, 79+83, 100, 126b and KH 91. The densest are HT 45a and 45b, which carry three, and which contain the sequence **\*318 · \*305** twice, always in that order and separated by a divider: **\*318 · \*305 2** and **\*318 · \*305 7**. That is the signature of a fixed pair.

### 3. The two dossiers compared

| commodity class | aromatics dossier                                | *305 dossier                |
|-----------------|--------------------------------------------------|-----------------------------|
| foodstuffs      | 29 documents (expected 5.1), <b>p &lt; 0.001</b> | 4 (expected 1.8), p = 0.100 |
| oil             | 15 (expected 2.1), <b>p &lt; 0.001</b>           | 2 (expected 0.7), p = 0.162 |
| people          | 4 (expected 1.3), p = 0.042                      | **3 (expected 0.5), p =     |

| commodity class              | aromatics dossier           | *305 dossier                |
|------------------------------|-----------------------------|-----------------------------|
|                              |                             | 0.011**                     |
| metals                       | 3 (expected 0.2), p = 0.001 | 1 (expected 0.1), p = 0.081 |
| livestock, textiles, vessels | 0                           | 0                           |

And the cleanest separation, which is one of form rather than of company:

|                   | fractions | whole units | % fractional |
|-------------------|-----------|-------------|--------------|
| aromatics dossier | 52        | 44          | **54%**      |
| *305 dossier      | 1         | 19          | **5%**       |

Of sixty-eight documents in the first and twenty-four in the second, **six overlap** (HT 23a, 33, 89, 100, KH 2, KH 57). Two dossiers of the same archive, counted in two different ways.

#### 4. The arithmetic that resolves \*305

The first hypothesis was that \*305 designated a material distributed to people. The arithmetic of two tablets says otherwise.

**HT 100** reads: VIR+KA 58 | \*304+PA [figure not preserved] | TI+A 12 | KI 2 | **\*305 5** | VIR+\*313a 16 | **KU-RO 97**. The five surviving figures sum to \*\*93\*\*. The missing figure would therefore be \*\*4\*\*, and the total would balance exactly.

**HT 94a**, of the same type and structure and with no losses, reads: VIR+[?] 62 | \*86 20 | TI+A 7 | VIR+\*313b 18 | TA 4 | **KU-RO 110**. Those figures sum to \*\*111\*\*, balancing within the one-unit margin usual in these accounts.

And HT 94a is what decides the case, because its personnel total includes **\*86 and TA, which are bare signs and not qualified VIR**. In this type of tablet the total of people sums categories written with single signs alongside the qualified men. On HT 100 the equivalent positions are occupied by KI and \*305.

**\*305 is therefore counted within a total of people**. It is not a commodity appearing beside a personnel account: it is one of the categories that account sums.

That resolves every property of the dossier at once. Whole numbers, because people are not divided. No fractions. No foodstuffs, livestock or textiles, because it is not a commodity. Association with qualified VIR, because it belongs to the same lists. And the fixed pair \*318 · \*305 would be a designation of two elements, as VIR+KA is a man plus a qualifier.

#### 5. The material candidates, and why they fall

Before the arithmetic, a list of materials was assembled that would satisfy the observed constraints — assigned to people, counted in whole units, escaping the accounting of wool, livestock and basic rations. It is set out here because three of the seven fall on corpus evidence rather than on plausibility, and that is worth recording.

| candidate                                          | verdict                                                                                                                                                                                          |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bronze or copper under a <i>ta-ra-si-ja</i> system | <b>falls:</b> Linear A has no metal logogram. Of those Linear B uses (AES, AUR, ARG) none occurs; only *171, twice, unidentified. An industry in bronze without a sign for bronze does not stand |
| aromatic resins, myrrh, compounded spices          | <b>falls:</b> these would belong to the aromatics dossier, and the two dossiers share six documents of ninety-two                                                                                |
| alum                                               | <b>weakened:</b> its destinations are leather and dye, and the dossier never co-occurs with textiles (0 of 24 documents)                                                                         |
| red ochre and pigments; faience paste              | not testable: no identified sign for either, and nothing in the data distinguishes them                                                                                                          |
| beeswax                                            | compatible, but subject to the same objection as resins: wax is a perfumery input, and the dossiers are separate                                                                                 |
| salt                                               | compatible and not excluded                                                                                                                                                                      |

One correction belongs here. An earlier version of this analysis held that the dossier “is weighed”, on the strength of the double mina on HT 45b. That was wrong: the double mina occurs on six documents and on none of them measures \*305. It measures PA-JA on HT 41a, KU-\*321 with grain on ZA 18a, KU-TU-KO-RE on ZA 8, cyperus and figs on HT 94a, and figs on HT 100. On HT 45b the figure “8 double minas” stands on a line separate from \*305, which carries 7 without a unit. The claim is withdrawn, and with it every candidate that depended on weight.

## 6. The point of contact between the dossiers

The two blocks touch at two points, wine and the QA<sub>2</sub> family, and the second is the more informative.

| ligature                    | dossier     | quantities           |
|-----------------------------|-------------|----------------------|
| QA <sub>2</sub> +PU         | aromatics   | 10, 1, 1/16          |
| QA <sub>2</sub> +RE         | aromatics   | 10, 1, 1/4           |
| <b>QA<sub>2</sub>+RE+PU</b> | <b>*305</b> | <b>**10, 1 1/4**</b> |

Three ligatures of the same base sign, distributed between two dossiers, and the third is the combination of the first two. If QA<sub>2</sub> marks a class of product, the two sectors use different

varieties of the same substance, and one of them additionally uses a compound form. Its quantities are comparable to or greater than those of its components, as would be expected.

#### **6 bis. The four personnel tablets are already described, and read as feasts**

The structure of the tablets used here is published. Montecchi (“Planning a Feast in Neopalatial Crete”) analyses HT 27a, 89, 94a and 100 as lists divided into two sections, the first recording personnel and the second three or four agricultural commodities in the same order, and reads them as records of feast preparation, by comparison with the Mycenaean feasting records studied by Palaima. That interpretation is not tested here and is not incompatible with anything above: a category of personnel counted for a feast is still a category of personnel. It does mean that the two-section structure should not be presented as an observation of this paper, and it is not.

#### **6 ter. A datum bearing on \*306**

Uchitel has proposed, in a paper devoted to the question, that \*306 is the Linear A sign for woman. His argument is typological in the first place: the absence of a distinction between signs for men and women is unparalleled in any comparable logographic-phonetic script, and he adds the palaeographic observation that the prototype of the Linear B MUL is attested in Cretan Hieroglyphic (\*004) and absent from the Linear A repertoire. The evidence he adduces is the occurrence of \*306 with numbered recipients against agricultural commodities, the pairing of roundels and tablets in which \*306 occupies the position of DI, and the reconstruction of HT 94a as men under sa-ra<sub>2</sub>, eleven women under \*318-\*306 and eleven children, with a 1:1 ratio of women to children as in the Pylian Aa and Ab series. He is also aware that \*306 occurs within syllabic sequences, and reads one of them, \*306-KI-TA<sub>2</sub>, as a candidate for the Minoan word for woman written with a determinative; a distributional count of syllabic against logographic use therefore does not decide the question either way.

One datum can be added, and it bears on the Pylian parallel, which is the principal support. In the Aa and Ab series the identification of groups by sex and age rests on the fixity of the rations. Measured across the six tablets Uchitel himself tabulates for the barley-figs-wine sequence (HT 27a, 30 twice, 89, 94a, 99a, 100), the ratios are not fixed: figs to barley runs from 0.38 to 1.00 and wine to barley from 0.06 to 2.18. **There is no fixed ration scale in this corpus.** That does not refute his reading of HT 94a, since a scale might be internal to a single tablet, but it removes the feature on which the Pylian comparison depends.

### **7. Discussion**

What is established. Class Ω of the 2019 classification corresponds to a dossier that can be recovered by distributional measurement; that dossier is distinct from the aromatics dossier of the same archive, sharing six of ninety-two documents; the two differ most sharply in how they count, 54% fractional against 5%; and \*305 enters a total of people on HT 100, with HT 94a showing that bare signs belong in such totals.

What is not established. What \*305 designates. The reading offered is one of function — a category of personnel — and not of value. Which category, the corpus does not say, and the

same holds for \*318, for the pair the two form, and for TU and ZU, which extend beyond Haghia Triada and are frequent single signs whose membership of the dossier is the weakest part of the block.

A limitation of the same kind as elsewhere in this work. The block was grown by hypergeometric co-occurrence, which controls how many documents each sign occurs in but not how many logograms each document carries; the documents of the aromatics block are richer in logograms than the corpus average, and the same check should be run here. It has not been.

What this suggests for the archive. Two dossiers of the same villa, one counted in fractions and dealing in oil and aromatics, the other in whole units and dealing in people, sharing a commodity family between them. If that is right, the question the archive poses is not only what Haghia Triada produced but **how many kinds of personnel its administration distinguished**, given that eleven forms of qualified VIR are attested and that several bare signs enter personnel totals. That question has not, so far as I know, been asked with numbers, and it is where I intend to go next.

### Data

Classification: Montecchi (2019), *Contare a Haghia Triada*, Incunabula Graeca CVII, table 4. Corpus: GORILA (Godart and Olivier 1976-1985) in the form of the LinearA Explorer (Hogan 2022), with sign verification in SigLA. Scripts: the kuro package (Acedo 2026d).

### Declaration of assistance

Corpus processing, the tests and the first draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

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